Name : _____ Branch : _____ SNU ID : _____

1. Temporal Cohesion means

(a) Cohesion between temporary variables
(b) Cohesion between local variables
(c) Cohesion with respect to time
(d) Coincidental cohesion

2. Functional Cohesion means
(a) Operations are part of single functional task and are placed in same procedures
(b) Operations are part of single functional task and are placed in multiple procedures
(c) Operations are part of multiple tasks
(d) None of the above

3. When two modules refer to the same global data area, they are related as
(a) External coupled
(b) Data coupled
(c) Content coupled
(d) Common coupled

4. The relationship of data elements in a module is called
(a) Coupling
(b) Cohesion
(c) Modularity
(d) None of the above

5. Which one of the following can be said for a sequence diagram?
(a) It is used to model the behavior of a single object when many use cases are executed
(b) It is used to model the behavior of a several objects when a single use case is executed
(c) It is used to model the behavior of a single object when a single use case is executed
(d) It is used to model the behavior of a several objects when many use cases are executed

6. Which one of the following is true for unary association?
(a) It is defined between multiple classes
(b) It is defined on exactly one object
(c) It is defined between multiple objects of the same class
(d) It is defined between multiple objects of different classes

7. When two classes are associated, which one of the following can be said about the links existing between the objects of the corresponding classes?
(a) A link exists between any two pair of objects belonging to the two classes
(b) Links between objects usually get dynamically formed and resolved as the execution proceeds
(c) Links between objects are predefined during compile time
(d) Links between objects can form during program execution, but none of the links gets dissolved

8. Consider the following programming situation. A certain class C1 has a method whose parameter list includes an object of class C2 as a parameter. Which one of the following types of relations can be said to exist between the classes C1 and C2?
(a) Inheritance
(b) Aggregation
(c) Composition
(d) Dependency

9. Draw the class diagram for Tic-Tac-Toe Game Software. **(1 Mark)**

10. Consider the following scenario of a Super Market and **draw the sequence diagram** for the "Select Winner List" Use Case. **(3 Marks)**

A super market needs to develop a software that would help it to automate a scheme that it plans to introduce to encourage regular customers. In this scheme, a customer would have first register by supplying his/her residence address, telephone number, and the driving license number. Each customer who registers for this scheme is assigned a unique *customer number* (CN) by the computer. A customer can present his CN to the checkout staff when he makes any purchase.

In this case, the value of his purchase is credited against his CN. At the end of each year, the supermarket intends to award surprise gifts to 10 customers who make the highest total purchase over the year.
Also, it intends to award a 22 caret gold coin to every customer purchase exceeded Rs. 10,000/-. The entries against the CN are reset on the last day of every year after the prize winners' lists are generated.

11. Draw the Class Diagram: **(3 Marks)**

Some instructors are professors, while others have job title adjunct professor. Departments offer many courses, but a course may be offered either by a single department or at most jointly by two departments. Courses are taught by instructors, who may teach up to three courses. Instructors are assigned to one departments. But, at times instructors are given joint appointment in two departments. One of the instructors of a department also serves as the head of the department.